P1 For completeness, we start by deriving

(a) $$dp^\alpha = -\frac{\mu_0 q^2 c}{6\pi} w^\beta w_\beta \, dx^\alpha \quad (1)$$ first.

(this is not required for the solution of the problem)

4-momentum $p^\alpha = \left(\frac{\mathcal{E}}{c}, \vec{p}\right) \Rightarrow dp^\alpha = \left(\frac{d\mathcal{E}}{c}, d\vec{p}\right)$

we have $dp^\alpha = A\, dx^\alpha = A(c\,dt, d\vec{r}) = A(c\,dt, \vec{v}\,dt)$

Thus $\frac{d\mathcal{E}}{c} = A c\,dt$, $\frac{d\mathcal{E}}{dt} = c^2 A$.

$d\vec{p} = A\,\vec{v}\,dt$, $\frac{d\vec{p}}{dt} = \vec{v} A$

$$\Rightarrow \frac{d\vec{p}}{dt} = \frac{\vec{v}}{c^2}\frac{d\mathcal{E}}{dt}, \quad d\vec{p} = \frac{\vec{v}}{c}\frac{d\mathcal{E}}{c}.$$

In the rest frame $(\vec{v}=0)$ $d\mathcal{E} = \frac{\mu_0 q^2}{6\pi c} a^2 dt =$

$$= \frac{\mu_0 q^2}{6\pi c}\left(-c^4 w^\beta w_\beta\right) dt$$

$$\Rightarrow dp^\alpha = \left(\frac{d\mathcal{E}}{c}, 0\right) = \left(\frac{\mu_0 q^2}{6\pi c^3}\left(-c^4 w^\beta w_\beta\right) c\,dt, 0\right)$$

$$= -\frac{\mu_0 q^2 c}{6\pi} w^\beta w_\beta \left(dx^0, 0\right)$$

Arbitrary frame

$$dp^\alpha = -\frac{\mu_0 q^2 c}{6\pi} \; w^\beta w_\beta \; dx^\alpha$$

$$w^\beta w_\beta = \left(\frac{\gamma^4}{c^3} \vec{v}\cdot\vec{a}, \; \frac{\gamma^4}{c^4}(\vec{v}\cdot\vec{a})\vec{v} + \frac{\gamma^2}{c^2}\vec{a} \right) \cdot \left(\; , - \; \right)$$

$$= \frac{\gamma^8}{c^6}(\bar{v}\cdot\bar{a})^2 - \left\{ \frac{\gamma^8}{c^6}(\bar{v}\cdot\bar{a})^2 \frac{v^2}{c^2} + \frac{\gamma^4}{c^4} a^2 + 2\frac{\gamma^6}{c^6}(\bar{v}\cdot\bar{a})^2 \right\}$$

$$= \frac{\gamma^8}{c^6}(\bar{v}\cdot\bar{a})^2 \underbrace{\left(1 - \frac{v^2}{c^2}\right)}_{+\gamma^{-2}} - \frac{\gamma^4}{c^4} a^2 - 2\frac{\gamma^6}{c^6}(\bar{v}\cdot\bar{a})^2 =$$

$$= -\frac{\gamma^6}{c^6}(\bar{v}\cdot\bar{a})^2 - \frac{\gamma^4}{c^4} a^2 = -\frac{\gamma^6}{c^6} v^2 a^2 - \frac{\gamma^4}{c^4} a^2 + \frac{\gamma^6}{c^6}(\bar{v}\times\bar{a})^2$$

$$(\bar{v}\cdot\bar{a})^2 = v^2 a^2 - (\bar{v}\times\bar{a})^2$$

$$w^\beta w_\beta = -\frac{\gamma^4}{c^4} a^2 \underbrace{\left(1 + \gamma^2 \frac{v^2}{c^2}\right)}_{1 + \frac{v^2/c^2}{1 - v^2/c^2} = \frac{1}{1 - v^2/c^2} = \gamma^2} + \frac{\gamma^6}{c^6}(\bar{v}\times\bar{a})^2 =$$

$$= -\frac{\gamma^6}{c^4}\left\{ a^2 - \frac{(\bar{v}\times\bar{a})^2}{c^2} \right\}.$$

$$dp^\alpha = \frac{\mu_0 q^2}{6\pi} \frac{\gamma^6}{c^3}\left[\vec{a}^2 - \frac{(\bar{v}\times\bar{a})^2}{c^2} \right] dx^\alpha$$

$$dp^\alpha = \left(\frac{d\mathcal{E}}{c}, d\vec{p}\right)$$

$$\frac{d\mathcal{E}}{c} = \frac{\mu_0 q^2}{6\pi} \frac{\gamma^6}{c^3}\left[a^2 - \frac{(\bar{v}\times\bar{a})^2}{c^2}\right] c\,dt$$

$$d\vec{p} = \frac{\mu_0 q^2}{6\pi} \frac{\gamma^6}{c^3}\left[a^2 - \frac{(\bar{v}\times\bar{a})^2}{c^2}\right] \vec{v}\,dt$$

$$d\mathcal{E} = \frac{\mu_0 q^2}{6\pi c} \gamma^6 \left[a^2 - \frac{(\bar{v}\times\bar{a})^2}{c^2}\right] dt$$

Observe that $d\vec{p} = \frac{\mu_0 q^2}{6\pi c} \gamma^6 \left[a^2 - \left(\frac{\bar{v}\times\bar{a}}{c}\right)^2\right] \frac{\vec{v}}{c^2} dt =$

$$= \frac{\vec{v}}{c^2} d\mathcal{E} \quad \text{(ok)}$$

Now, for a particle moving in external EM field

$$mc \frac{du^\alpha}{ds} = q F^{\alpha\beta} u_\beta$$

EM 4-tensor

4-acceleration $w^\alpha = \frac{du^\alpha}{ds}$

$$w_\alpha = \frac{du_\alpha}{ds} = \frac{q}{mc} F_{\alpha\beta} u^\beta$$

$$\Rightarrow w^\alpha w_\alpha = \left(\frac{q}{mc}\right)^2 F^{\alpha\beta} u_\beta F_{\alpha\gamma} u^\gamma$$

$$\underline{dp^\delta} = -\frac{\mu_0 q^2 c}{6\pi}\left(\frac{q}{mc}\right)^2 F^{\alpha\beta} u_\beta F_{\alpha\gamma} u^\gamma \underline{dx^\delta}$$

$$F^{\alpha\beta} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

$$F_{\alpha\beta} = \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

$\alpha = 0$: $\quad mc\, W^\alpha = q\, F^{\alpha\beta}\, u_\beta \quad , \quad u_\beta = \gamma\left(1, -\vec{v}/c\right)$

$$\Rightarrow mc\, W^0 = q\, F^{0\beta}\, u_\beta = q\left(-\frac{\vec{E}}{c}\right)\left(-\gamma\,\frac{\vec{v}}{c}\right) = \frac{\gamma q}{c^2}\,\vec{E}\cdot\vec{v}$$

$\alpha = 1,2,3$: $\quad mc\, W^\alpha = q\, F^{1\beta}\, u_\beta = q\left(F^{10} u_0 + F^{12} u_2 + F^{13} u_3\right)$

$$= \frac{q}{c} E_x\, u_0 + q\left(-B_z\, u_2 + B_y\, u_3\right) =$$

$$= \frac{\gamma q}{c} E_x + \frac{\gamma q}{c}\left(B_z\, v_y - B_y\, v_z\right)$$

$$\Rightarrow mc\, W^1 = \frac{\gamma q}{c}\left[\vec{E} + \left(\vec{v}\times\vec{B}\right)_x\right].$$

$$u_0 = \gamma \ , \ u_2 = -\frac{\gamma}{c} v_y \ , \ u_3 = -\frac{\gamma}{c} v_z$$

$$mc\, w^1 = \frac{\gamma q}{c} E_x + \frac{\gamma q}{c}\left(B_z v_y - B_y v_z\right)$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{x}\left(v_y B_z - v_z B_y\right)$$

$$\Rightarrow mc\, w^1 = \frac{\gamma q}{c}\left(E_x + \left(\vec{v} \times \vec{B}\right)_x\right)$$

$$mc\, \vec{w} = \frac{\gamma q}{c}\left(\vec{E} + \vec{v} \times \vec{B}\right)$$

Then $dp^\alpha = -\frac{\mu_0 q^2 c}{6\pi} \frac{1}{(mc)^2}\, mc\, w^\beta\, mc\, w_\beta\, dx^\alpha$

$$mc\, w^\beta\, mc\, w_\beta = \left(\frac{\gamma q}{c^2} \vec{v} \cdot \vec{E}, \frac{\gamma q}{c}\left(\vec{E} + \vec{v} \times \vec{B}\right)\right) \cdot$$

$$\cdot \left(\frac{\gamma q}{c^2} \vec{v} \cdot \vec{E}, -\frac{\gamma q}{c}\left(\vec{E} + \vec{v} \times \vec{B}\right)\right) =$$

$$= \left(\frac{\gamma q}{c^2}\right)^2 \left(\vec{v} \cdot \vec{E}\right)^2 - \left(\frac{\gamma q}{c}\right)^2 \left(\vec{E} + \vec{v} \times \vec{B}\right)^2$$

$$= -\left(\frac{\gamma q}{c}\right)^2 \left\{\left(\vec{E} + \vec{v} \times \vec{B}\right)^2 - \frac{1}{c^2}\left(\vec{v} \cdot \vec{E}\right)^2\right\}$$

$$\Rightarrow \quad dp^\alpha = + \frac{\mu_0 q^2 c}{6\pi} \frac{1}{(mc)^2} \left(\frac{\gamma q}{c}\right)^2 \left\{ \left(\vec{E} + \vec{v}\times\vec{B}\right)^2 - \frac{1}{c^2}\left(\vec{v}\cdot\vec{E}\right)^2 \right\} dx^\alpha$$

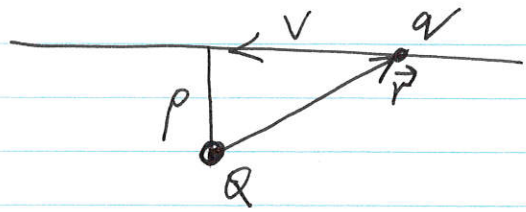
$$= \frac{\mu_0 q^4 \cancel{\gamma}}{6\pi m^2 c^3} \left[\frac{\left(\vec{E} + \vec{v}\times\vec{B}\right)^2 - \frac{1}{c^2}\left(\vec{v}\cdot\vec{E}\right)^2}{1 - v^2/c^2} \right] dx^\alpha$$

$$\Rightarrow \frac{d\mathcal{E}}{c} = \frac{\mu_0 q^4}{6\pi m^2 c^3} \left[\quad \right] c\,dt$$

$$\frac{d\mathcal{E}}{dt} = \frac{\mu_0 q^4}{6\pi c m^2} \frac{\left(\vec{E} + \vec{v}\times\vec{B}\right)^2 - \frac{1}{c^2}\left(\vec{v}\cdot\vec{E}\right)^2}{1 - v^2/c^2}.$$

$$\frac{d\vec{p}}{dt} = \frac{\vec{v}}{c^2}\frac{d\mathcal{E}}{dt}$$

(b)



$$\vec{r}(t) = x\,\hat{x} + \rho\,\hat{y} = vt\,\hat{x} + \rho\,\hat{y}$$

$$\vec{v} = -v\,\hat{x}$$

$$\Delta\mathcal{E} = \int_{-\infty}^{\infty} \frac{d\mathcal{E}}{dt}\,dt$$

$$\vec{B} = 0 \quad , \quad \vec{E} = \frac{Q}{4\pi\varepsilon_0}\frac{\hat{r}}{r^2} = \frac{Q\,\vec{r}}{4\pi\varepsilon_0 r^3} = \frac{Q}{4\pi\varepsilon_0}\frac{vt\,\hat{x} + \rho\,\hat{y}}{(\rho^2+v^2t^2)^{3/2}}$$

$$\frac{d\mathcal{E}}{dt} = \frac{\mu_0 q^4}{6\pi c m^2}\,\frac{E^2 - \frac{1}{c^2}(\vec{E}\cdot\vec{v})^2}{1 - v^2/c^2} =$$

$$= \frac{\mu_0 q^4}{6\pi c m^2}\,\gamma^2\left(\frac{Q}{4\pi\varepsilon_0}\right)^2 \frac{v^2t^2 + \rho^2 - \frac{1}{c^2}(v^2t)^2}{(\rho^2+v^2t^2)^3}$$

$$= \frac{\mu_0 q^4 Q^2}{96\pi^3\varepsilon_0^2 c m^2}\,\gamma^2\left\{\frac{1}{(\rho^2+v^2t^2)^2} - \frac{v^4t^2}{c^2(\rho^2+v^2t^2)^3}\right\}$$

$$\Delta\mathcal{E} = \frac{\mu_0 q^4 Q^2}{96\pi^3\varepsilon_0^2 c m^2}\,\frac{\gamma^2}{v}\int_{-\infty}^{\infty} d(vt)\left\{\frac{1}{(\rho^2+v^2t^2)^2} - \frac{v^2}{c^2}\frac{(vt)^2}{(\rho^2+v^2t^2)^3}\right\}$$

$$\int_{-\infty}^{\infty}\frac{dx}{(x^2+\rho^2)^2} = \frac{\pi}{2\rho^3} \quad , \quad \int_{-\infty}^{\infty}\frac{dx\,x^2}{(\rho^2+x^2)^3} = \frac{\pi}{8\rho^3}$$

$$\Delta\mathcal{E} = \frac{\mu_0 q^4 Q^2}{96\pi^3\varepsilon_0^2 c m^2}\,\frac{\pi}{8\rho^3}\,\frac{\gamma^2}{v}\left[4 - \frac{v^2}{c^2}\right] = \frac{\mu_0 q^4 Q^2}{768\pi^2\varepsilon_0^2 c m \rho^3 v}\,\frac{4c^2 - v^2}{c^2 - v^2}$$

(c) $\vec{E} = 0 \Rightarrow \dfrac{d\mathcal{E}}{dt} = \dfrac{\mu_0 q^4}{6\pi c m^2} \dfrac{(\vec{v} \times \vec{B})^2}{1 - v^2/c^2}$

$\vec{v} \perp \vec{B} \Rightarrow (\vec{v} \times \vec{B})^2 = v^2 B^2$

Total energy emitted $\Delta\mathcal{E} = \int_0^T \dfrac{d\mathcal{E}}{dt}\, dt$

$$\Delta\mathcal{E} = \frac{\mu_0 q^4}{6\pi c m^2} \frac{v^2 B^2}{1 - v^2/c^2} \cdot T \quad (T = \text{period})$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi R}{v}, \quad \omega = \text{cyclotron frequency}$$

$$T = \frac{2\pi m}{qB\sqrt{1 - v^2/c^2}}$$

$$\Delta\mathcal{E} = \frac{\mu_0 q^4}{6\pi c m^2} \frac{v^2 B^2}{1 - v^2/c^2} \frac{2\pi m}{qB\sqrt{1 - v^2/c^2}} =$$

$$= \frac{\mu_0 q^3}{3cm} \frac{v^2 B}{(1 - v^2/c^2)^{3/2}}$$

$$\frac{d\vec{p}}{dt} = q\,\vec{v}\times\vec{B} \quad , \quad \vec{B} = B\hat{z}$$

$$\begin{cases} \dfrac{dp_x}{dt} = q\,v_y B \\[2ex] \dfrac{dp_y}{dt} = -q\,v_x B \end{cases}$$

But $\vec{p} = \dfrac{\vec{v}\,\mathcal{E}}{c^2}$ and $\mathcal{E} = \text{const}$

$$\begin{cases} \dfrac{dv_x}{dt} = \dfrac{qc^2B}{\mathcal{E}}\,v_y \\[2ex] \dfrac{dv_y}{dt} = -\dfrac{qc^2B}{\mathcal{E}}\,v_x \end{cases} \Rightarrow d(v_x + i v_y) = -i\omega\,(v_x + i v_y)$$

$$\omega = \frac{qc^2B}{\mathcal{E}}$$

$$\Rightarrow v_x + i v_y = v_0\, e^{-i\omega t}$$

$v_x^2 + v_y^2 = v_0^2 = v^2$ = constant speed in x-y plane.

$$v_x = v_0 \cos\omega t \Rightarrow x = x_0 + \frac{v_0}{\omega}\sin\omega t$$

$$v_y = -v_0 \sin\omega t \Rightarrow y = y_0 + \frac{v_0}{\omega}\cos\omega t$$

$\Rightarrow R = \dfrac{v_0}{\omega}$ = cyclotron radius.

$$\omega = \frac{qc^2B}{\mathcal{E}} = \frac{qc^2B}{c^2 p/v} = \frac{qvB}{p} = \frac{qB}{m}\sqrt{1 - v^2/c^2}$$

$$p = \frac{mv}{\sqrt{1 - v^2/c^2}}$$

P2 $\vec{A} = A_0 \hat{e}\, e^{i(\vec{k}\cdot\vec{r}-\omega t)}$

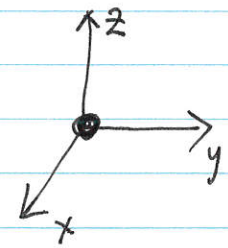
$$\vec{E}_{in} = -\frac{\partial \vec{A}}{\partial t} = i\omega A_0 \hat{e}\, e^{+i(\vec{k}\cdot\vec{r}-\omega t)}$$

$$m\ddot{\vec{r}} = q\vec{E} \quad , \vec{d} = q\vec{r} \Rightarrow \ddot{\vec{d}} = \frac{q^2}{m}\vec{E}$$

$$\vec{B}_{rad} = \frac{\mu_0}{4\pi r}\ddot{\vec{d}} \times \hat{n} = \frac{\mu_0 q^2}{4\pi r m}\vec{E}_{in} \times \hat{n}$$

$$\vec{B}_{in} = \vec{\nabla}\times\vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ 0 & A & 0 \end{vmatrix} = -\hat{x}\,\partial_z A + \hat{z}\,\partial_x A$$

let $\vec{k} = k\hat{x}$, $\hat{e} = \hat{y} \Rightarrow \vec{k}\cdot\vec{r} = kx$



$$\partial_z A = 0 \quad , \partial_x A = A_0(ik)\, e^{i(kx-\omega t)}$$

$$\vec{B}_{in} = i k A_0 e^{i(kx-\omega t)}\, \hat{z}$$

$$\vec{E}_{in} = i\omega A_0 e^{i(kx-\omega t)}\, \hat{y}$$

$$\vec{B}_{rad} = \frac{\mu_0 q^2}{4\pi r m}\, i\omega A_0 e^{i(kx-\omega t)}\, \underbrace{\hat{y}\times\hat{n}}$$

for $\hat{n} = \hat{x}$: $\hat{y}\times\hat{x} = -\hat{z}$

$$\vec{B}_{TOT} = \vec{B}_{in} + \vec{B}_{rad} = i k A_0 e^{i(kx-\omega t)}\, \hat{z}\left(1 - \frac{\mu_0 q^2}{4\pi r m}\underbrace{\left(\frac{\omega}{k}\right)}_{c}\right)$$

$$\mu_0 c = \frac{1}{\varepsilon_0 c}$$

the <u>minus</u> sign describes destructive interference.